The Medical Expansion, Life-Expectancy and Endogenous Directed Technical Change

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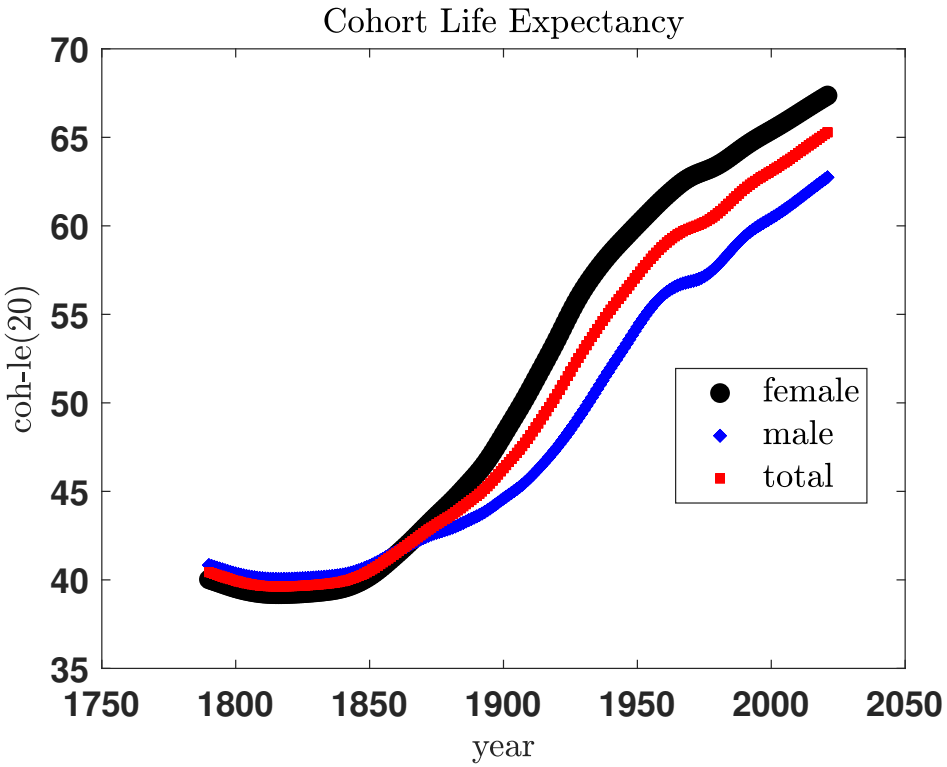
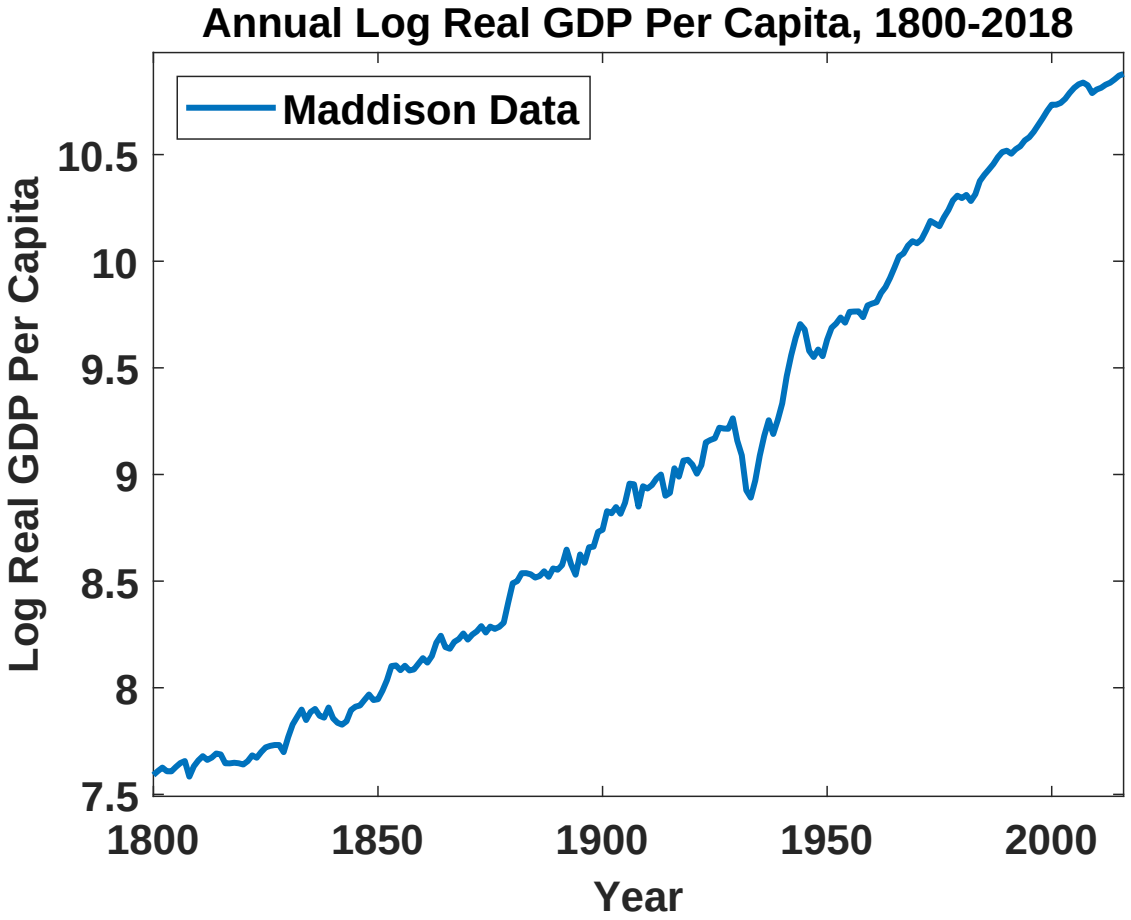
Remaining *Cohort* Life Expectancy at Age 20 in U.S.



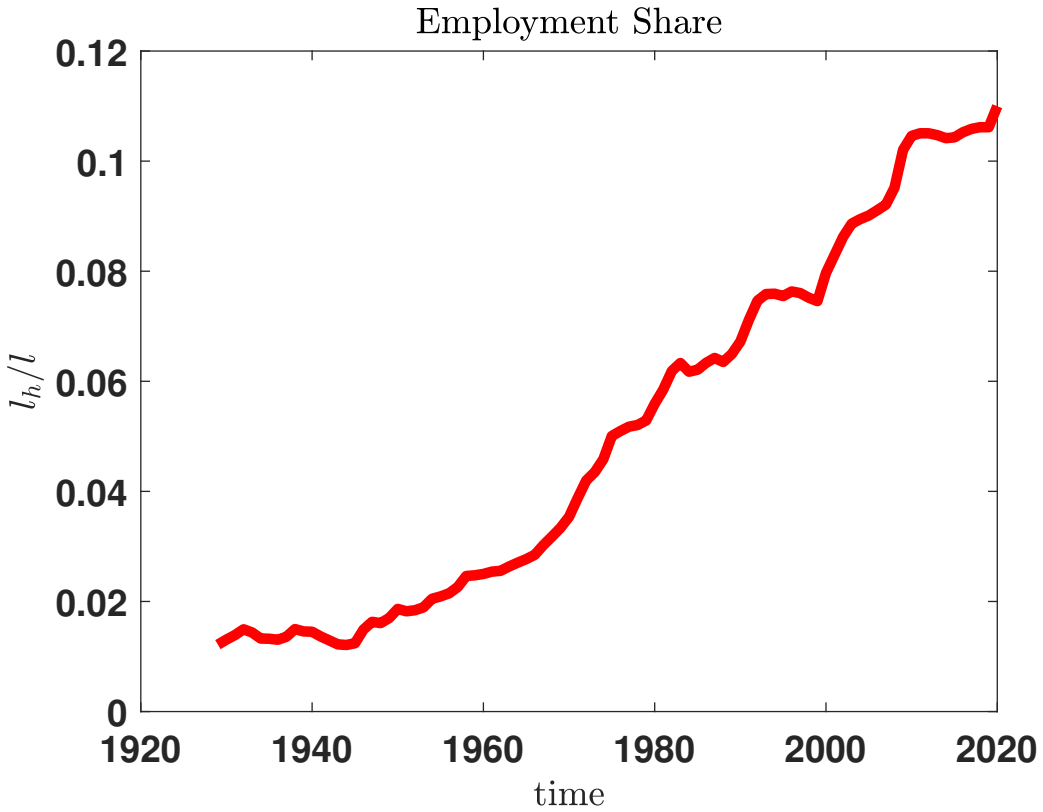
- Flat adult life expectancy at age 20 until about 1840. Then take-off.
- Source: Historical Life Expectancy Data (Haines, Hacker 2010), Human Life-Table Database, Human Mortality Database.

Per Capita Income Growth



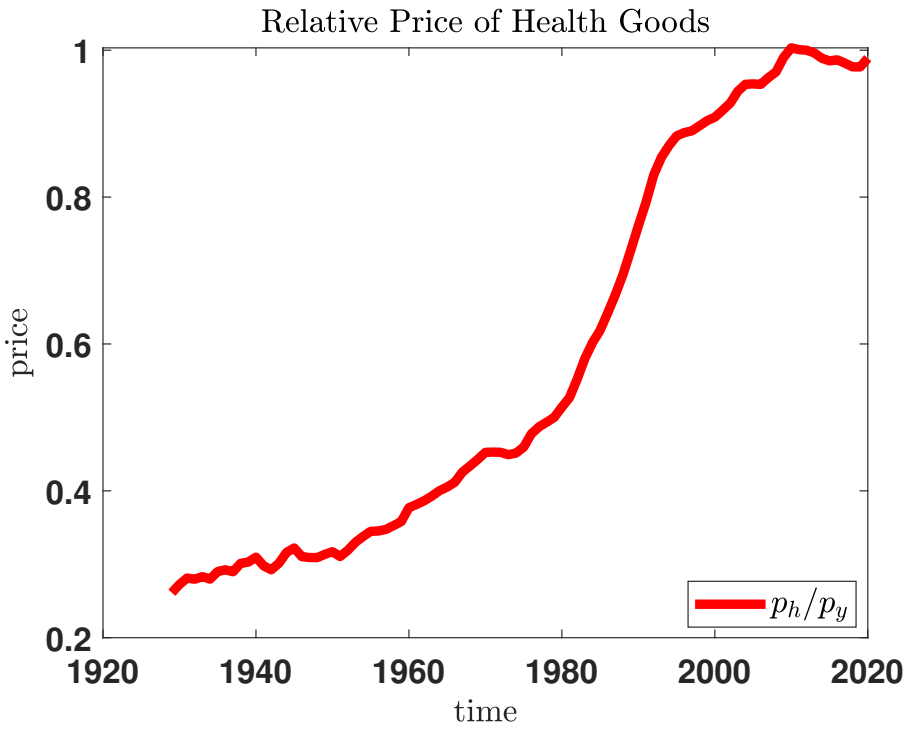
- Per capita income (log scale) started increasing in about 1820
- Roughly constant growth at about 2% annually since

Health Employment Share



- Share of workforce employed in health sector ↑ since WW.II.
- Similar trends for health expenditure share and health output share.
- Penicillin first developed in 1929. Widespread use since WW.II

Relative Price of Health Goods



- Increase of relative price of health goods & services
- Quality adjustment?

Motivation and Research Objective

- Facts: Three Phases of Health and Medical Development
  1. Life Expectancy at Age 20 flat until about 1840.
  2. Life Expectancy at Age 20 ↑ since about 1840.
  3. Emergence of Modern Health Sector 1920-40: Employment, Output, Expenditure Share ↑
- Objective: Quantitative theory, predict future, evaluate policies
- Building Blocks:
  1. Life Cycle: Diamond (1965)
  2. Endogenous Health Investment & Longevity: Grossman (1972)
  3. Endogenous Directed Technical Change: Aghion & Howitt (1992)

Modeling Approach

- Two-sector OLG model with endogenous technical change:
  - Households:
    - ⋆ 2-periods lived, endogenous survival to 2nd period.
    - ⋆ Choices: consumption-savings, health spending.
    - ⋆ Two health goods: basic hygiene & modern health services.
  - Firms:
    - ⋆ Two sectors: health goods & final goods
    - ⋆ Monopolistic competition in intermediate inputs ⇒ Profits
    - ⋆ Endogenous R&D: ⇒ higher quality intermediates ⇒ Profits.
    - ⋆ Endogenous income growth through technological ↑ in both sectors.
- Quantitative implementation: Calibration to initial conditions, broad trends in US data.

Main Mechanism (aka the "Story")

- Phase 1: Low productivity & Low Income ⇒ No Health Spending.
- Phase 2: Productivity growth in basic goods sector ⇒ Income ↑

  => Kick-off: Basic health spending ↑, life expectancy ↑.
- Phase 3: Further income ↑ & non-homotheticity in health spending

  => Health spending ↑

  => Redirection of technological progress to modern health sector.

  => Quality in modern health sector ↑, price of health goods ↑.

  => Convergence to interior BGP.

Results Today

- Construction & calibration of simple, illustrative model.
- Calibrated model results: replicate facts quantitatively (sort of).
- Health Policy reforms: not yet.

Outline

1. Introduction
2. Economic Model
3. Calibration
4. Results
5. Conclusion

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Model: Overview

Two-sector OLG model with endogenous technical change:

- Households:
  - 2-periods lived, endogenous survival to 2nd period.
  - Consumption-savings choice
  - Demand: basic food & hygiene goods & modern health goods.
- Firms and Technology:
  - Two final goods sectors: modern health goods & generic consumption goods (includes hygiene & food).
  - Both sectors: continuum of intermediate inputs. Imperfect substitution & monopolistic competition $\Rightarrow$ Profits
  - Endogenous R&D: $\Rightarrow$ higher quality of intermediates $\Rightarrow$ Profits.
  - Endogenous income growth through quality $\uparrow$ in both sectors.
- SOE: interest rate $R_t = R$ exogenous, constant.

Households: Utility and Choices

- Consumption-savings (c^o_{t+1}, s_t), health investment (i_t, i_{ht}, i_{ft}) given prices p_t, R
- Utility from old-age consumption and survival:

$$\psi(i_t)v(c^o_{t+1}) = \psi(i_t)\left(\frac{(c^o_{t+1})^{1-\sigma}}{1-\sigma} + b\right)$$

- No suicide condition: b sufficiently large (required if $\sigma \geq 1$).
- Survival probability increases in i_t:

$$\psi(i_t) = 1 - (1 + i_t)^{-\xi}.$$

- Health investment quasi-linear in basic, modern health goods:

$$i_t = i_{ht} + \left(\nu + i_{ft}\right)^{\zeta}$$

- Note that $\psi'(i_{ht} = i_{ft} = 0) < \infty$ but $u'(c_{t+1} = 0) = \infty$.
- Budget constraints:

$$c^y_t + p_t i_{ht} + i_{ft} + s_t := e_t + s_t = w_t + T_t := x_t$$
$$c^o_{t+1} = Rs_t$$

Analysis of Household Problem: Three Phases

Proposition

Suppose $\frac{x_0}{p_0}$ *is sufficiently low and that the sequence of prices & cash at hand* $\{p_t, x_t\}$ *satisfies:*

$$\frac{x_{t+1}}{p_{t+1}} > \frac{x_t}{p_t}.$$

Then there exist time thresholds $0 < T_1 < T_2 < \infty$ *such that*

1. *Phase 1:* $\forall t < T_1$*:* $i_t = i_{ft} = i_{ht} = 0$, $\psi(i_t) = \psi(0)$
2. *Phase 2:* $\forall t \in [T_1, T_2)$*:* $i_t = i_{ft} > 0$, $i_{ht} = 0$ *&* $\psi(i_t) > \psi(0)$. *Life expectancy* $\uparrow$*: better basic hygiene, no modern health sector.*
3. *Phase 3: For all* $t \geq T_2$ *we have* $i_{ft} > 0$ *&* $i_{ht} > 0$ *as well as* $\psi(i_t) > \psi(0)$. *Life expectancy* $\uparrow$*, also modern health goods* $\uparrow$.
4. *BGP w/ constant* $\frac{p \cdot i_h}{x} > 0$, $\frac{i_f}{x} = 0$, $\frac{s}{x} > 0$, $\frac{c}{x} > 0$ *&* $p > 0$.

Production Side: Final Goods Production Firms

- Perfectly competitive final goods producers with CRTS technology in both sectors $j \in \{f, h\}$:

$$y_{jt} = \left(\int_0^1 q_{jit}^{1-\alpha} y_{jit}^{\alpha} \right) l_{jt}^{1-\alpha}$$

- Firms take as given:
  - Quality q_{jit} and prices p_{jit} of intermediate goods
  - Prices of final goods and wages p_{jt}, w_{jt} in sector j.
- Choices: y_{jt}, l_{jt}, y_{jit}
- FOC's for y_{jit} delivers inverse demand function for intermediates:

$$p_{jit} = \alpha p_{jt} \left(\frac{q_{jit} l_{jt}}{y_{jit}} \right)^{1-\alpha}$$

Intermediate Inputs: Monopolistic Competition

- Each variety $i \in [0,1]$ is produced by a monopolist.
- Production function: $y_{jit} = k_{jit}$, full depreciation of capital k_{jit}.
- Firms take as given: inverse demand function & R.
- Profit maximization:

$$\pi_{jit} = \max_{k_{jit}} \left\{ \left[\alpha p_{jt} \left(\frac{q_{jit} l_{jt}}{k_{jit}} \right)^{1-\alpha} \right] k_{jit} - R k_{jit} \right\}$$

- Solution: constant markup over marginal cost R, positive profits:

$$p_{jit} = \frac{1}{\alpha} R > R, \quad \pi_{jit} = \frac{1-\alpha}{\alpha} R k_{jit} > 0$$

Firms: Aggregating the Production Sector

- From intermediate goods producers' FOC: For all $i \in [0,1]$,

$$\frac{k_{jit}}{q_{jit}} = \frac{k_{jt}}{q_{jt}},$$

where $q_{jt} = \int_0^1 q_{jit} di$ & $k_{jt} = \int_0^1 k_{jit} di$.

- Aggregation in each sector:

$$y_{jt} = k_{jt}^{\alpha} \left(q_{jt} l_{jt}\right)^{1-\alpha}$$

- Distribution of income:

$$p_{jt} y_{jt} = \left[(1-\alpha) + \alpha^2 + \alpha(1-\alpha)\right] p_{jt} y_{jt} = w_t l_{jt} + R k_{jt} + \pi_{jt}$$

R&D Production & Technological Progress

- R&D entrepreneur per variety i: resources z_{jit} on innovation.
- Probability of successful innovation:

$$\phi(z_{jit}; l_{jt}, q_{jit-1}) = \min\left[\varphi\left(\frac{z_{jit}}{\lambda q_{jit-1} l_{jt}}\right)^{\gamma}, 1\right]$$

- Successful innovation: quality improvement $\lambda > 1$ so that $q_{jit} = \lambda q_{jit-1}$.
- Successful innovator: one period monopolist for i: Profits π_{jit}.
- R&D entrepreneur's problem:

$$\max_{z_{jit}} \{\pi_{jit} \cdot \phi(z_{jit}; l_{jt}, q_{jit-1}) - z_{jit}\}$$

  Solution $z_{jit} = \Phi(R, p_{jt})\lambda q_{jit-1} l_{jt}$.
- Varieties i w/ unsuccessful innovations: quality $q_{jit} = q_{jit-1}$, randomly selected entrepreneur eats profits π_{jit}.

Firms: Aggregation of R&D & Economic Growth

- Since $\frac{z_{jit}}{\lambda q_{jit-1} l_{jt}} = \Phi(R, p_{jt})$ constant across i:

$$\mu_{jt} = \varphi \left(\frac{z_{jit}}{\lambda q_{jit-1} l_{jt}} \right)^{\gamma} = \varphi \left(\Phi(R, p_{jt}) \right)^{\gamma}$$

- Quality improvements as engine of growth:

$$q_{jt} = \mu_{jt} \lambda q_{jt-1} + (1 - \mu_{jt}) q_{jt-1}$$

- Growth rate in sector j:

$$g_{jt} = \frac{q_{jt}}{q_{jt-1}} = 1 + (\lambda - 1)\mu_{jt}.$$

Price & Quality of Health Goods

- Good f is the numeraire: $p_{ft} = 1$ for all t.
- Relative price of health goods per health efficiency unit i_{ht}:

$$p_{ht} =: p_t = \left(\frac{q_{ft}}{q_{ht}}\right)^{1-\alpha}$$

- Relative price, quality adjustment:

$$p_t \frac{q_{ht}}{q_{ft}} = \left(\frac{q_{ht}}{q_{ft}}\right)^{\alpha}$$

Balanced Growth Path (BGP) and Transition

- Interior BGP: quality (q_{ft}, q_{ht}), x_t, w_t, T_t grow at rate g.
- Constant prices $R, p_t = p$. Constant shares:

$$\frac{e_t}{x_t} = \frac{p_t i_{ht} + i_{ft}}{x_t} = \frac{p_t i_{ht}}{x_t} = \vartheta, \frac{s_t}{x_t} = 1 - \vartheta, \frac{c_{t+1}}{x_t} = R(1 - \vartheta)$$

- BGP with interior share $\vartheta = \frac{e}{x} \in (0, 1)$ exists iff $\sigma = 1 + \xi$.
- Why? FOC w.r.t. $\vartheta_t = \frac{e_t}{x_t}$ equates marginal benefit of health spending (longer life) to cost (reduced consumption):

$$\max_{\vartheta_t} \left(1 - \frac{1}{(1 + i_t(\vartheta_t x_t))^{\xi}}\right) \left(\frac{(Rx_t(1 - \vartheta_t))^{1-\sigma}}{1 - \sigma} + b\right)$$

- For (c_{t+1}, e_t) to grow at same rate: $\sigma = 1 + \xi$.

Transition to BGP

- State of the economy $(q_{ht-1}, q_{ft-1}, n_t, s_{t-1})$
- Given state (& $R_t = R$): static equilibrium, determine p_t (or $\frac{l_{ft}}{l_{ht}}$).
- Assumption $\sigma = 2$, thus $\xi = 1$: closed-form for interior $\vartheta_t \Rightarrow$

  demand for health goods $\Rightarrow$ update of state $\Rightarrow (n_{t+1}, s_t)$.
- Relative price p_t determines $l_{ft}, l_{ht}, \mu_{ft}, \mu_{ht}$.
- Update of state: $\Rightarrow (q_{ht}, q_{ft})$.

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Extensions for Quantitative Analysis

- Utility from consumption also when young (otherwise implausible asset flows):

$$(1-\beta)u(c_t^y) + \beta\psi(i_t)v(c_{t+1}^o) = (1-\beta)\frac{{c_t^y}^{1-\sigma}}{1-\sigma} + \beta\psi(i_t)\left(\frac{{c_{t+1}^o}^{1-\sigma}}{1-\sigma} + b\right)$$

- Labor intensive health sector: $\alpha_h = 0.22, \alpha_f = 0.33$. (Acemoglu and Guerrieri 2008).
- Differential improvement factors: λ_j
- Key optimality conditions (& requirement for BGP) qualitatively unchanged (still need $\Rightarrow \sigma = 1 + \xi$). Currently $\sigma = 2$.
- Nonstandard time constraint slows down transition of labor across sectors. Size governed by elasticity ϵ. Currently: $\epsilon = 2$.

$$\left(l_{ft}^{1+\frac{1}{\epsilon}} + l_{ht}^{1+\frac{1}{\epsilon}}\right)^{\frac{1}{1+\frac{1}{\epsilon}}} = 1$$

Questions we Ask of the Model

- Basic Question 1: Can the model replicate basic empirical facts?
  - Life expectancy at age 20
  - Existence & size of modern health sector
  - Relative price of health goods
- 40 year model periods: young 20-59, old 60-99
- 6 periods: 1820 (phase 1), $1860, 1900$ (phase 2), $1940, 1980, 2020$ (phase 3).
- Question 2: What quantitative role does modern health sector play in expansion of life expectancy.
- (Future) Question 3: (Optimal) role of government in health R&D.

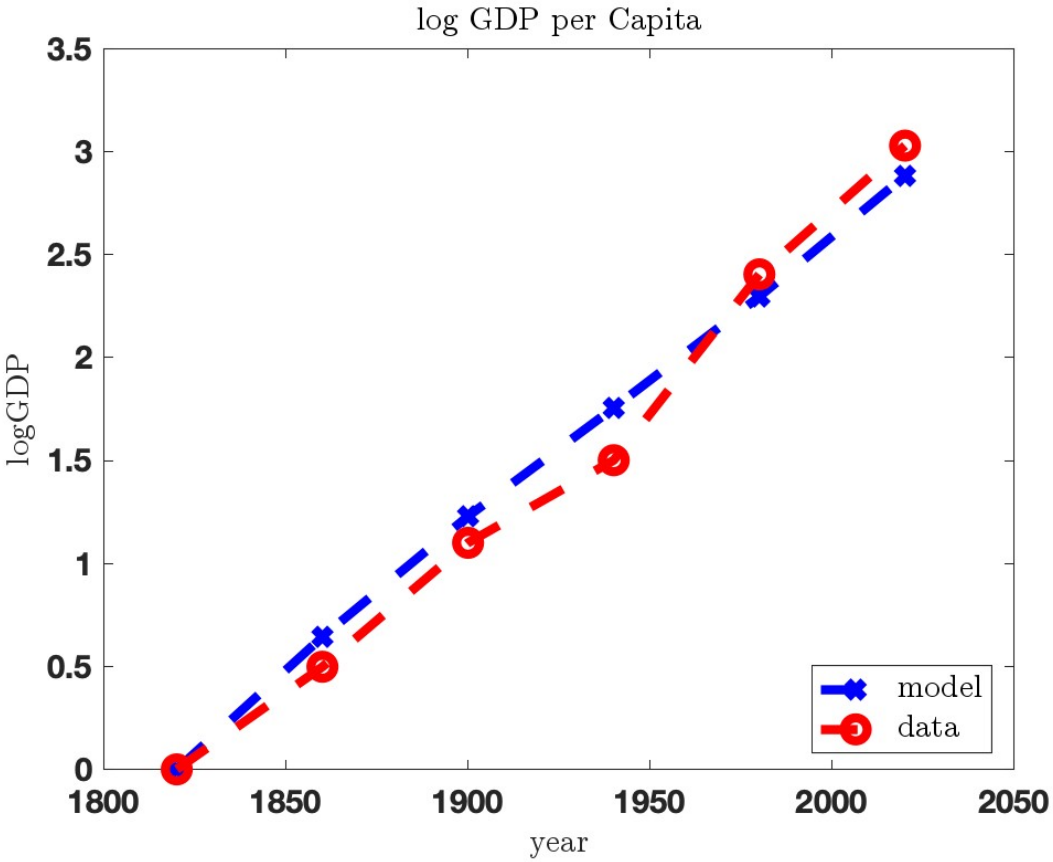
Calibration Strategy

- Broadly: pick parameters to get first two phases and timing of third phase right.
  - Value of life b: kick-off of basic health good spending
  - Initial quality gap: kick-off of modern health good spending
  - Minimum survival probability: adult remaining life expectancy of 40.2 years in 1790.
  - Growth factor λ_f: overall GDP growth
- IES $1/\sigma = 0.5$ standard. $\Rightarrow \xi = 1$.
- Growth factor λ_h: relative growth of modern health sector
- Evaluate the model wrt to performance of third phase.

Parameters

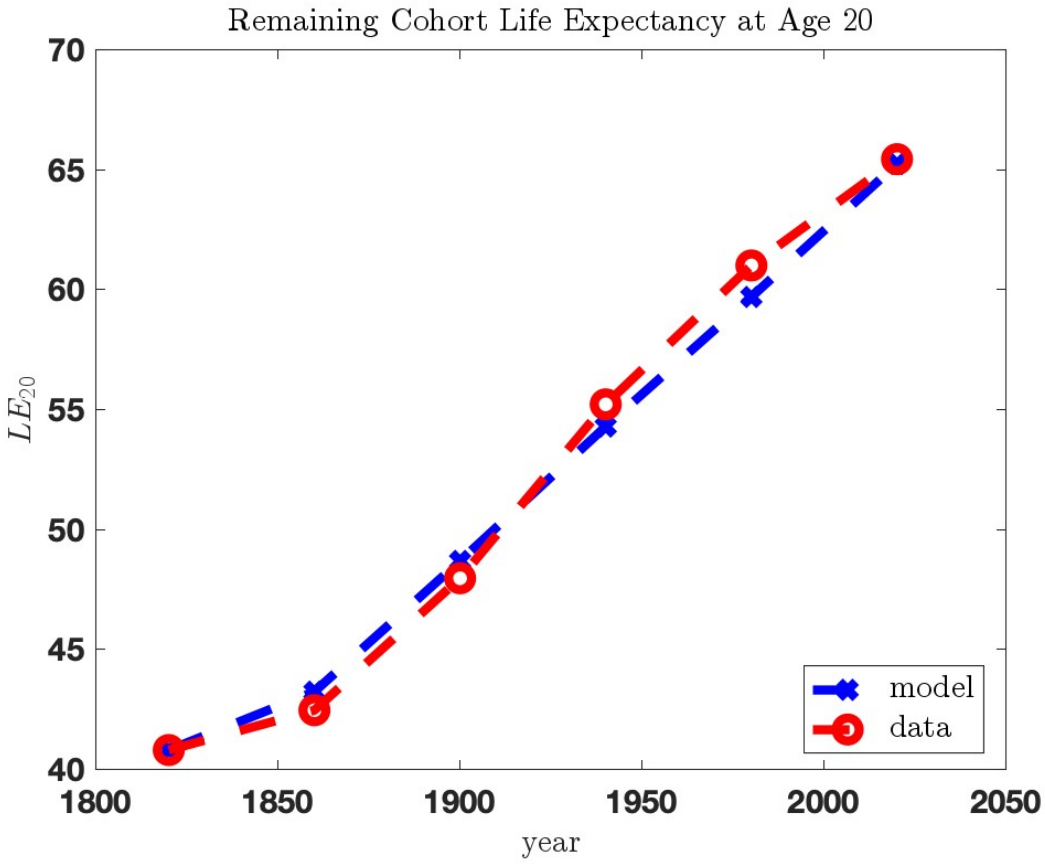
SOE	
Real Rate R-1	1.5 ($\approx$ 1 % annually)
Initial Condition	
Quality gap $\frac{q_{h0}}{q_{f0}}$	0.089
Households	
Discount Factor $\beta/(1-\beta)$	0.085 ($\approx$ 0.94 annually)
Value of Life b	130
IES $1/\sigma$	0.5
Tail parameter, survival function ξ	1
Min. surv. prob. at i = 0, ν^{ζ}	0.021
Scale parameter, modern health investment η	1
Labor mobility ϵ	2
Firms	
Capital elasticities $[\alpha_f\ ,\ \alpha_h]$	[0.33,0.2]
Growth factor $[\lambda_f\ ,\ \lambda_h]$	[120,100]
Innovation probability, curvature $[\gamma_f\ ,\ \gamma_h]$	[0.5,0.5]
Innovation probability, scale $[\varphi_f\ ,\ \varphi_h]$	[0.5,0.5]

Comparison to Data: Log GDP per Capita



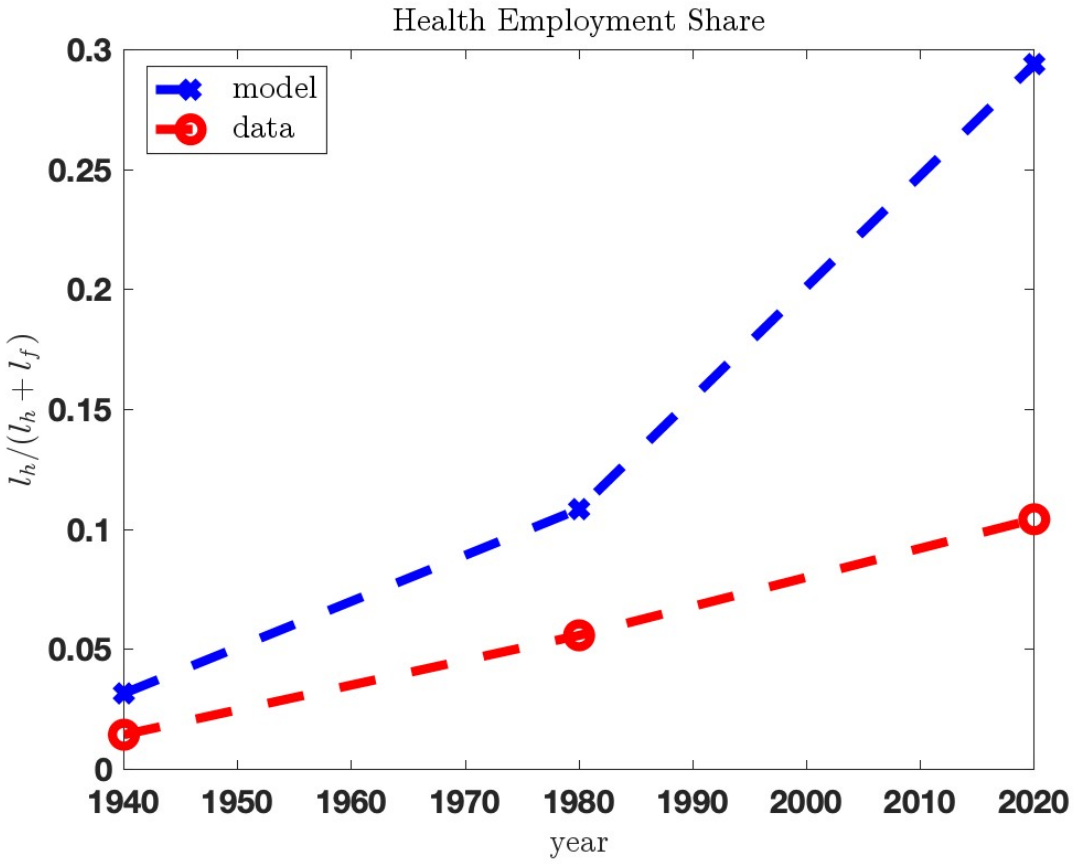
- Comparison looks good (easy to match)

Transition: Life Expectancy at Age 20



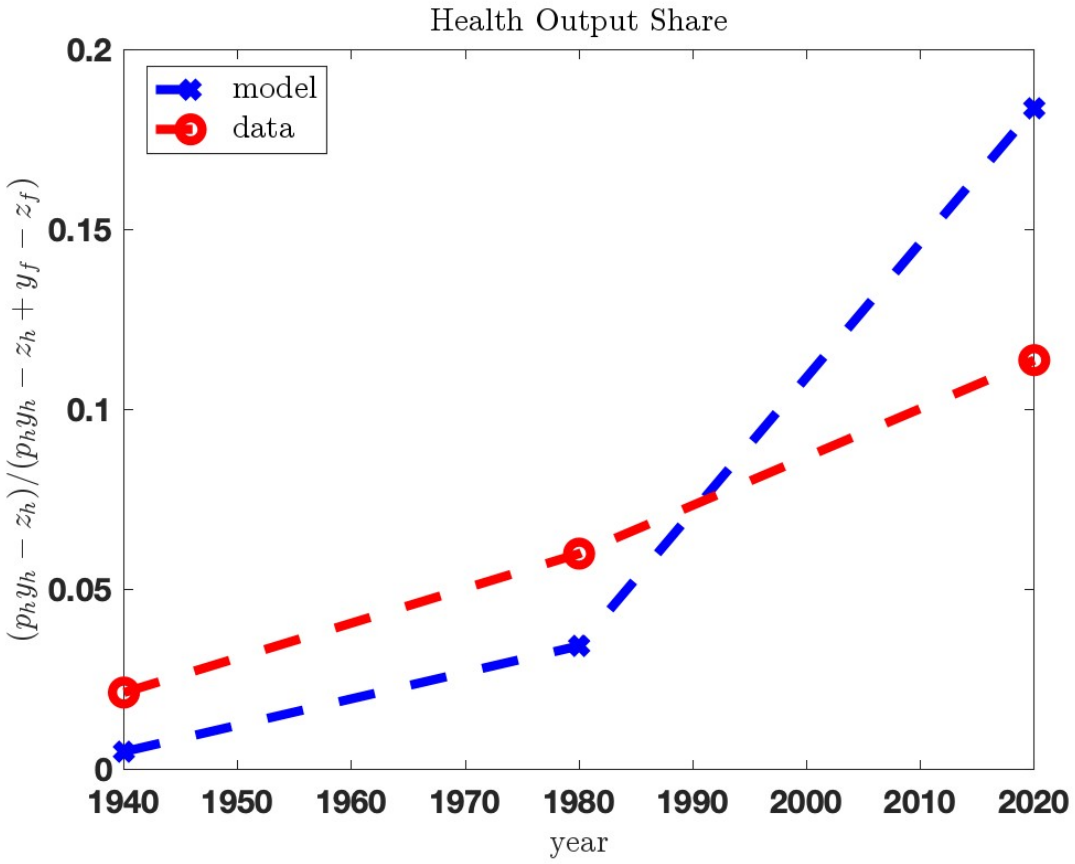
- Constant LE prior to kick-off, then increasing.

Comparison to Data: Health Employment Share



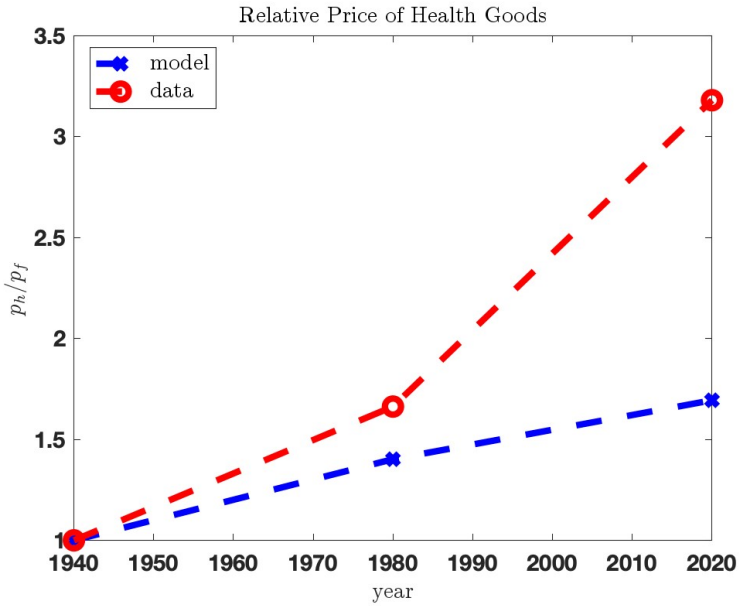
- Matches increase qualitatively, but too rapid quantitatively

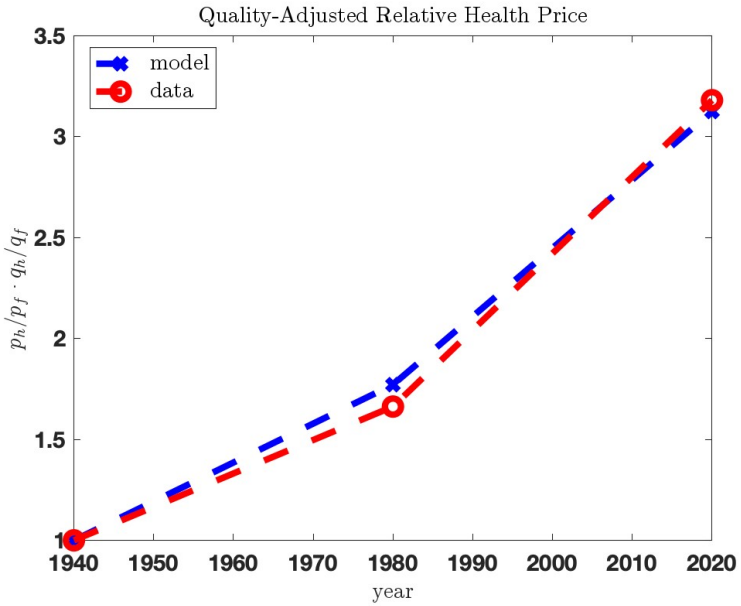
Comparison to Data: Health Output Share



- Matches increase qualitatively, but too rapid quantitatively

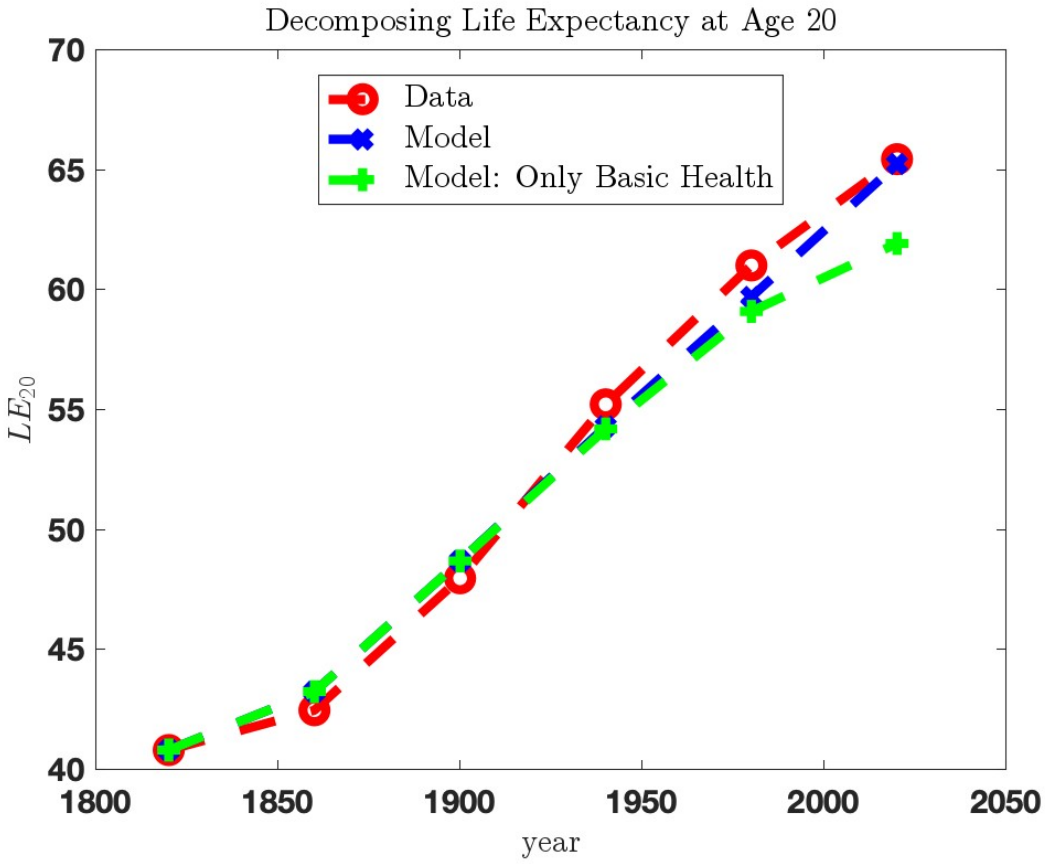
Comparison to Data: Relative Price





- Matches increase qualitatively
- With "re-adjusted" for quality close

Decomposing Life Expectancy at Age 20



- Growing contribution of modern health after 2nd kickoff

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Conclusion: What We Have Done So Far...

Endogenous growth model with a health sector generating...

- ... kick-off of adult life expectancy and (later) modern medicine
- ... positive trend of health spending share
- ... positive trend of health employment, R&D spending shares
- ... increasing relative price of health
- ....continuously increasing life-expectancy in 20-th century

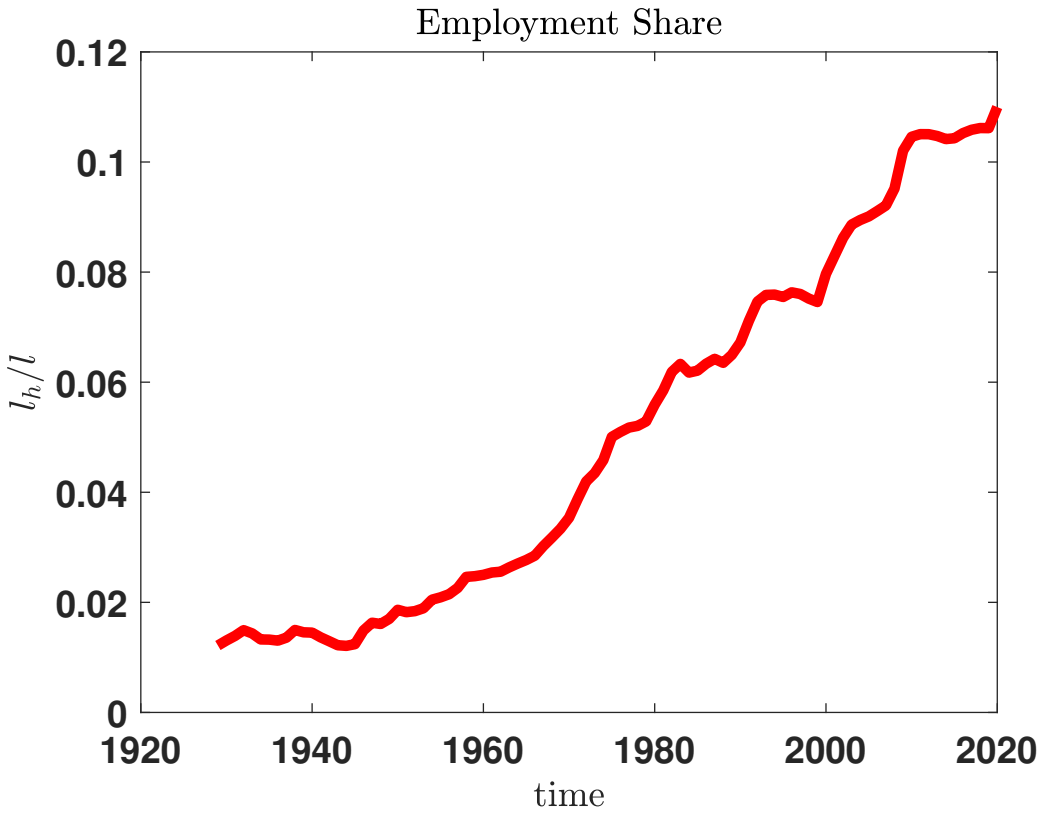
Conclusion: Next Step and Outlook

- Quantitative evaluation: reforms to health care & public R&D policies
- Model elements:
  - Life Cycle Model
  - Explicit model of health accumulation and frailty
  - consumption, savings, health investment, & endogenous retirement
  - household heterogeneity in life expectancy
  - Private & social insurance: health insurance & social security

THANK YOU FOR ATTENDING AND LISTENING

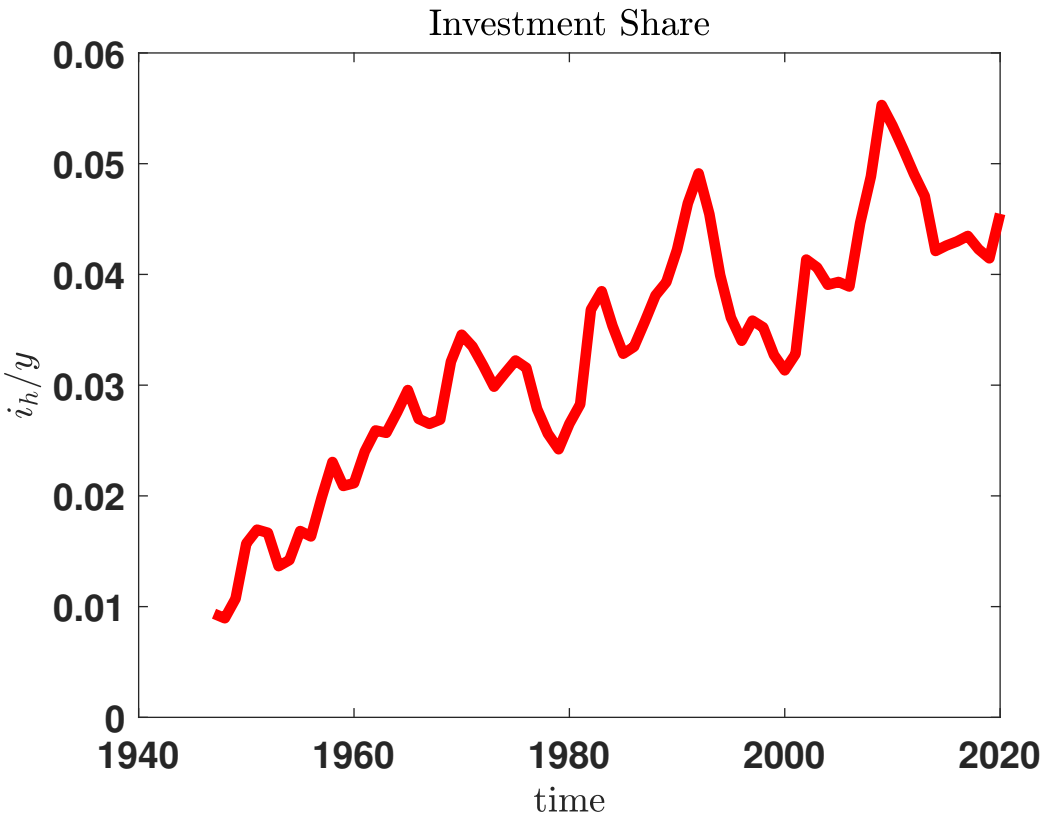
APPENDIX

Health Employment Share



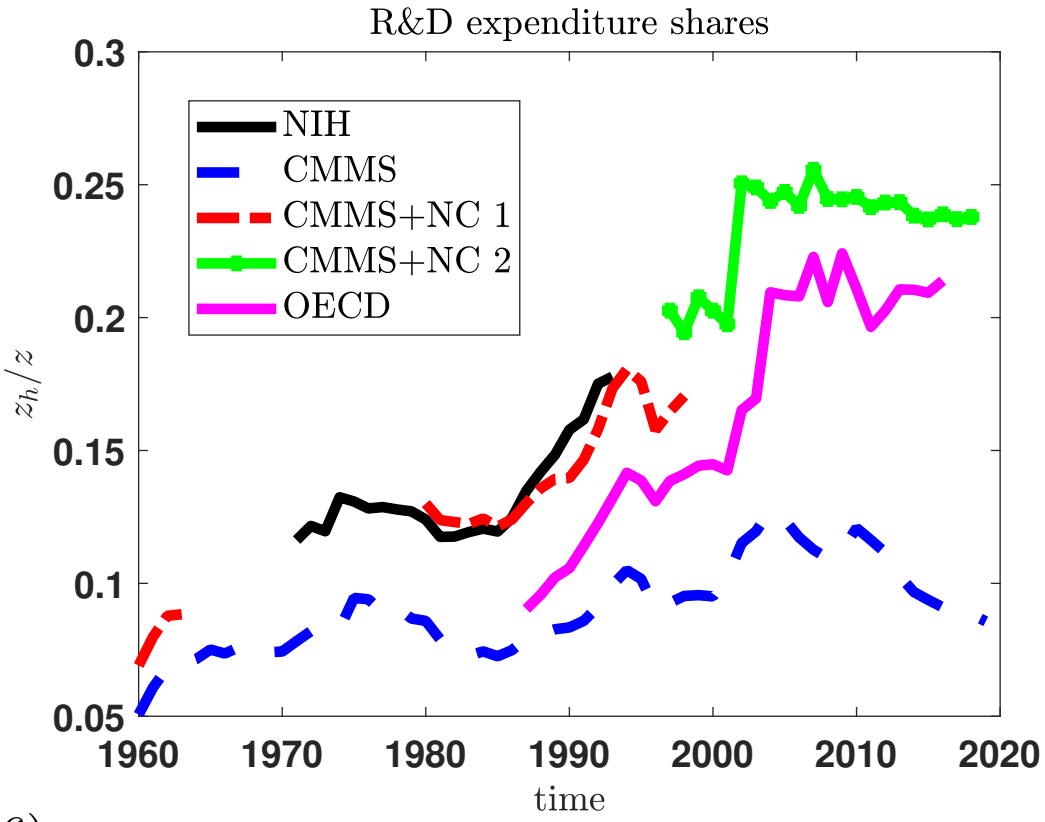
- Employment share ↑ since WW.II.

Investment Share



- Investment share ↑
- Data limitation

R&D Expenditure Share



Source: Jones (2016)

- R&D expenditure share ↑
- Data limitation

Related Literature

- Aghion-Howitt meets Grossman meets Diamond

  Diamond (1965), Grossman (1972), Aghion and Howitt (1992, 1998)

- Life expectancy, human capital & technological progress

  Cervellati & Sunde (2005), Hejkal, Ravikumar & Vandenbroucke (2022)

- Normative analyses of optimal health & R&D spending shares

  Hall and Jones (2007), Jones (2004, 2016)

- Reasons for growth of health spending

  Anderson et al. (2003), Fonseca et al. (2013), Zhao (2014), Hollingsworth et al. (2022)

- Health spending, R&D & feedback

  Frankovic and Kuhn (2018a,b), Böhm et al. (2018)

- Demographic change & directed technical change

  Ludwig et al. (2012), Heer & Irmen (2014), Acemoglu & Restrepo (2017, 2021)